

OIS-019 — System Model

Status: Foundational

Authority: Implementation

Depends on:

- OIS-000 through OIS-018
 - OAS-000 through OAS-031
 - OCS-000 through OCS-032
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Purpose

This specification defines the canonical System Model of Organodynamics.

The System Model provides the implementation-independent view of the complete executable ecosystem.

It does not introduce new constitutional concepts.

It organizes existing constitutional, architectural, and implementation responsibilities into one coherent operational model.

The System Model SHALL describe the whole.

It SHALL not redefine its parts.

1. Implementation Objective

The objective of the System Model is to provide a complete operational map from constitutional principles to executable behavior.

The Model SHALL preserve:

Constitutional hierarchy.

Architectural organization.

Implementation boundaries.

Operational relationships.

Verification pathways.

System comprehensibility.

2. Canonical Layers

Every conforming ecosystem SHALL be understandable through the following hierarchy.

```
graph TD; A["Constitution (Truth)"] --> B["Architecture (Organization)"]; B --> C["Implementation (Behavior)"]; C --> D["Runtime (Execution)"]; D --> E["Infrastructure (Operation)"];
```

Constitution (Truth)

↓

Architecture (Organization)

↓

Implementation (Behavior)

↓

Runtime (Execution)

↓

Infrastructure (Operation)

Each layer SHALL remain accountable to the layers above it.

3. Canonical Components

The System Model SHALL identify the major executable responsibilities.

Typical components include:

Place Runtime.

Identity Resolution.

Schema Registry.

Event Store.

Conversation Engine.

Knowledge Engine.

Decision Engine.

Federation Engine.

Verification Services.

Recovery Services.

Reference Model.

Reference Implementation.

Additional components MAY be introduced without changing the overall System Model.

4. Information Flow

Constitutional information SHALL flow through the ecosystem according to the following conceptual progression.

Identity

↓

Events

↓

Relationships

↓

Knowledge

↓

Reasoning



Operational optimizations SHALL preserve equivalent constitutional progression.

5. Operational Relationships

Components SHALL cooperate through explicit Service Contracts.

Relationships SHALL preserve:

Identity.

Historical continuity.

Behavioral equivalence.

Operational independence.

Relationship direction SHALL remain explicit.

6. Runtime View

The Runtime SHALL execute:

Conversations.

Knowledge construction.

Decision processes.

Federation.

Reflection.

Learning.

Operational execution SHALL remain constitutionally accountable.

7. Infrastructure View

Infrastructure SHALL provide operational capabilities including:

Storage.

Networking.

Scheduling.

Deployment.

Monitoring.

Security.

Infrastructure SHALL remain constitutionally invisible.

8. Recovery View

The complete ecosystem SHALL remain reconstructable using:

Constitution.

Architecture.

Schemas.

Events.

Canonical identities.

Recovery SHALL preserve equivalent constitutional behavior.

9. Verification View

Verification SHALL evaluate the complete System Model.

Every canonical component SHALL remain:

Observable.

Replaceable.

Traceable.

Constitutionally accountable.

10. System Model Test

The System Model satisfies this specification when an independent engineering organization can reconstruct an equivalent operational ecosystem solely from the Constitutional, Architectural, and Implementation Specifications without consulting any implementation-specific documentation.

Conformance

An implementation conforms to this specification when every executable responsibility can be located within the System Model while preserving constitutional meaning, architectural intent, operational equivalence, and independent replaceability.

Implementation Principle

Individual engines explain behavior.

Individual specifications explain responsibilities.

The System Model explains the organism.

It exists so that future engineers may understand not merely how individual parts operate,

but why they cooperate,

how they evolve together,

and how the complete constitutional ecosystem remains alive even as every implementation within it changes.

End of Specification OIS-019